



Unifying Global and Local Scene Entities Modelling for Precise Action Spotting

Introduction

Method: UGL Architecture

- Global Environment Feature

- Global Environment Feature

- Local Scene Entities

- Temporal Reasoning

Datasets & Task Definition

Training & Evaluation

- Evaluation Metrics

Results

Conclusion

- **Domain:** Action spotting in sports videos (soccer, diving, gymnastics)

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- **Applications:**
 - ▶ Highlight generation
 - ▶ Video analytics and broadcasting automation
 - ▶ Downstream temporal retrieval

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Motivation for UGL

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- Two-phase approaches: heavy 3D backbones, multiple extractors
- End-to-end methods (e.g., E2E-Spot): lighter but still global-only
- **Need**: explicit modeling of both global and local scene elements

UGL Architecture

Global Environment Feature

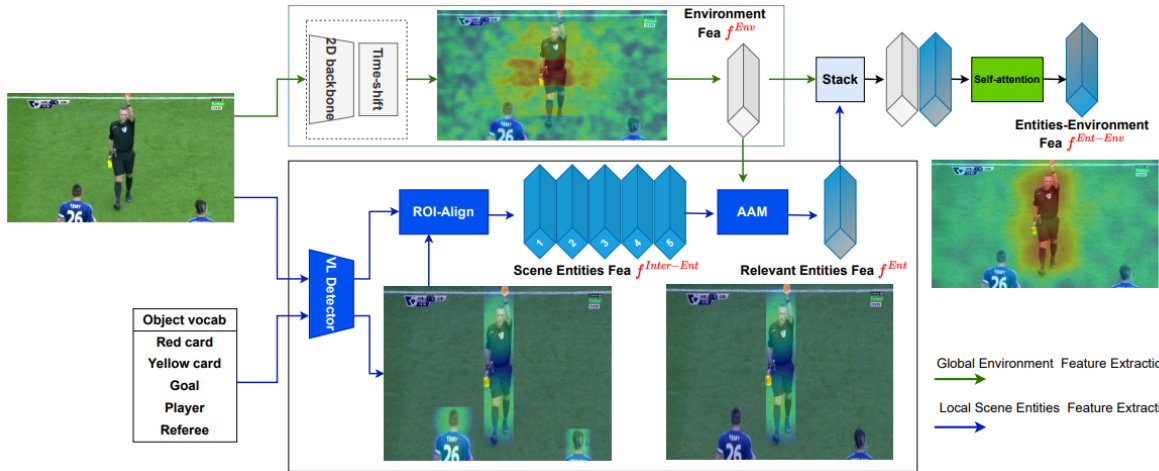


Fig. 2. The architecture of our proposed UGL module. Given a frame u , our UGL module concurrently extracts both the global environment feature f^{Env} and the local scene entities feature f^{Ent} .

- **Backbone:** RegNet-Y (efficient 2D CNN design)

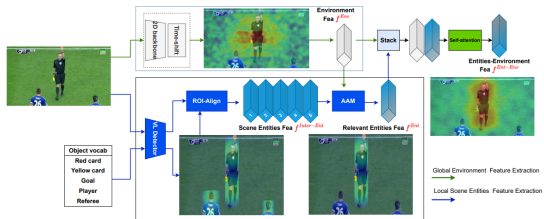


Fig. 2. The architecture of our proposed UGL module. Given a frame v_n , our UGL module concurrently extracts both the global environment feature f^{Env} and the local relevant entities feature f^{Ent} . Subsequently, it combines these features to produce a unified global-local entities-environment feature $f^{Ent-Env}$, acting as the unifying link between the local and global representations.

- **Backbone:** RegNet-Y (efficient 2D CNN design)
- **Temporal Integration:** Gate-Shift Module (GSM)
 - ▶ Shifts feature maps forward and backward along temporal axis
 - ▶ Shares information across frames
 - ▶ Lightweight alternative to 3D CNNs

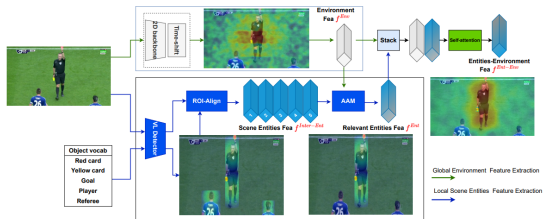


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Global Environment Feature

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- **Temporal Integration:** Gate-Shift Module (GSM)
 - ▶ Shifts feature maps forward and backward along temporal axis
 - ▶ Shares information across frames
 - ▶ Lightweight alternative to 3D CNNs
- **Output:** Global environment feature vector per frame capturing spatial and short-term temporal context

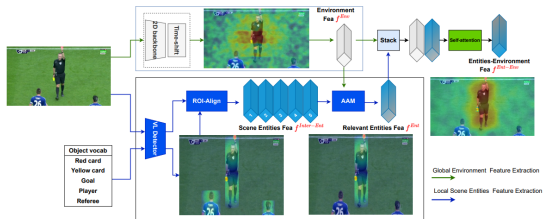


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Why GLIP?

Global features alone often ignore small but crucial entities (cards, ball, etc.), especially under class imbalance

- **GLIP**: Grounded Language-Image Pre-training (training-free detector)
- **Input**: Frame image + text prompt listing sports entities
 - ▶ *“ball, yellow card, red card, referee, ...”*
- **Mechanism**:
 - ▶ Language encoder processes text (BERT)
 - ▶ Visual encoder produces image proposals (Swin backbone)
 - ▶ Cross-modal multi-head attention aligns proposals with textual entities

Selecting Relevant Entities

Problem

Not all detected entities are equally relevant for the current action

AAM Solution

Takes both local entity features and global environment feature:

1. **Hard attention**: Select most promising entities
2. **Soft self-attention**: Fuse them into single compact representation

Output

Local relevant scene entities feature emphasizing objects crucial for the action

Fusion of Global and Local

Unified Representation

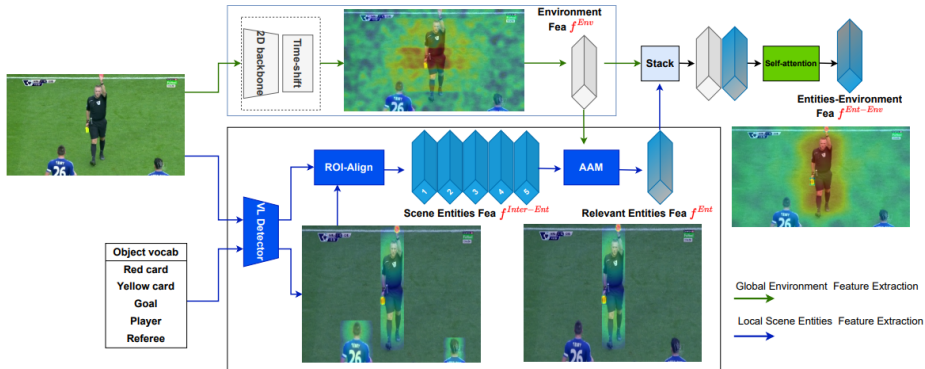


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Unified Representation

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- ▶ Global environment feature
- ▶ Local relevant entities feature

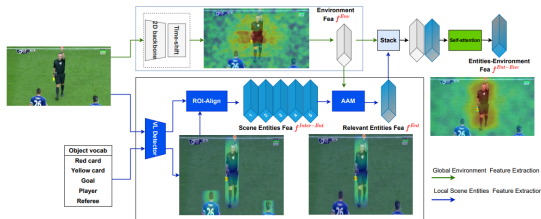


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Fusion of Global and Local

Unified Representation

1. Concatenate:

- ▶ Global environment feature
- ▶ Local relevant entities feature

2. Apply:

- ▶ Self-attention layer to model interactions
- ▶ Average pooling for compactness

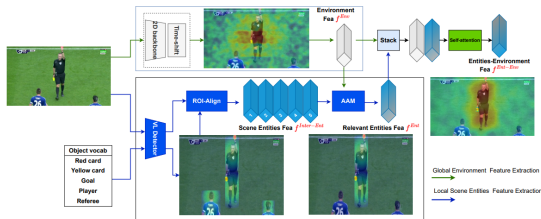


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Unified Representation

1. Concatenate:

- ▶ Global environment feature
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2. Apply:

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3. Output: Unified entities and environment representation per frame capturing both background context and key object semantics

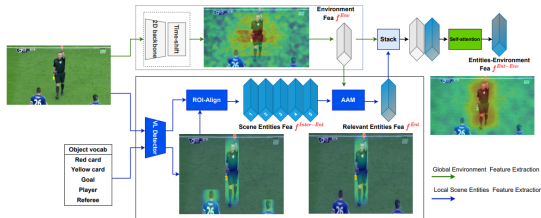


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Temporal Modeling

- Input: Sequence of fused frame features for T -frame snippet
- Model: 1-layer bidirectional GRU capturing semantic relationships
- Prediction: Per-frame class scores via fully connected + softmax

Inference with Sliding Window

- Slide window over entire video with overlap
- Merge frame-wise scores from GRU for each snippet
- Apply Non-Maximum Suppression (NMS) in time
- Output: Temporally precise event spots

Input

An untrimmed sports video and a set of action classes

Output

A *sparse* set of timestamps $\{(t_i, a_i)\}$, each with an action label, indicating when an event occurs

Evaluation

Mean AP over temporal tolerances, including tight tolerances (1–5 seconds) to measure precise spotting

SoccerNet-v2

- ~550 matches
- ~764 hours of video
- >300k labeled spots
- Strong clutter, camera changes

FineDiving

- 3,000 diving clips
- Phase transitions (entry, twist, etc.)

FineGym

- 5,374 gym performances
- 32 spotting classes
- Converted from action segments to timestamps

Focal Loss for Long-Tail Classes

The Problem

In SoccerNet-v2: background frames dominate; non-background classes $\approx 2\%$ of frames

Solution: Focal Loss

- Down-weights easy examples with high predicted probability
- Focuses learning on hard, misclassified, or minority-class examples
- Includes balancing parameter for class weighting
- Applied per frame over predicted multi-class outputs

Architecture

- RegNet-Y (ImageNet-1K init)
- GLIP-L (Swin-Large, no fine-tuning)
- Bidirectional GRU

Snippets

- Randomly sampled 100 frames

Data Augmentation

- Cropping, jittering, mixup
- Resize to height 224

Optimization

- AdamW optimizer
- Linear warm-up + cosine scheduling
- Single A100 80GB GPU
- 50–150 epochs per dataset

mAP

Mean Average Precision per class at fixed temporal tolerance

Average-mAP (A-mAP)

Area under mAP vs tolerance curve across range of tolerances

Tight Average-mAP (T-mAP)

Focuses on tight tolerances from 1 to 5 seconds, better reflects precise spotting capability

Top-1 Performance

UGL Leaderboard Performance

- **69.38% T-mAP — Top-1 rank**

Improvements over Key Baselines

- vs. E2E-Spot: +2.65% T-mAP
- vs. Baidu (two-phase): +19.82% T-mAP
- vs. Soares (two-phase): +1.57% T-mAP

Strong on Shown Actions

Large gains especially on events visible in frame; two-phase SOTA retains slight advantage only on unshown events

Consistent Improvements

FineDiving

- UGL consistently outperforms baselines
- vs. E2E-Spot (no optical flow):
- +2.4% mAP at selected tolerances

FineGym

- UGL consistently outperforms baselines
- vs. E2E-Spot (no optical flow):
- +1.3% mAP improvement

Across Setup Variants

All comparisons: pre-trained features, fine-tuned features, VC-Spot baseline, and E2E-Spot

Temporal Encoder & Loss

- GRU outperforms Transformer encoder with both cross-entropy and focal loss
- Focal Loss consistently beats cross-entropy, especially on long-tail classes

Local Entities & Fusion

- Fused feature (global + local via UGL) yields higher T-mAP than global-only
- Noticeable improvements on long-tail: penalty, red card, yellow-to-red

Explicit Entity Modeling Enables

- Highlight which objects (ball, card, player) drive a detection
- More interpretable reasoning vs. global black-box CNN + RNN

Visualizations in Paper

- Detection boxes showing GLIP outputs
- Attention maps focusing on relevant entities during events

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- Field of Teaching: **Artificial Intelligence, with Focus on Deep Learning**

General Information

- E-Mail: c.bergler@oth-aw.de
- Office Hours: Upon Request
- Internal/External Theses (Bachelor's/Master's) and AI-Projects possible at any time

- Action detection in sports videos: long-standing benchmarking problem
- GLIP-based vision-language grounding for weakly supervised object detection
- Gate-Shift Modules: efficient temporal modeling in 2D CNNs
- Focal Loss: addressing class imbalance in detection tasks

Further Questions?

